

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	57800	52.211302
1	[Git, Change, Create]	16910	15.274967
2	[Git, Commit]	11176	10.095390
3	[Git, Change, Delete]	8703	7.861505
4	[File, Git]	5776	5.217517
5	[Git, Change, Rename]	3315	2.994472
6	[Git, Tag]	1703	1.538336
7	[Author, Git, Person]	1261	1.139074
8	[Json, Key]	668	0.603411
9	[Json, Value, Scalar]	603	0.544696
10	[Committer, Git, Person]	370	0.334225
11	[NPM, Dependency]	338	0.305319
12	[Type, TS, Primitive]	291	0.262863
13	[Type, TS, Declared]	277	0.250217
14	[TS, ExternalDeclaration]	215	0.194212
15	[Type, TS, Literal]	136	0.122850
16	[Json, Value, Object]	133	0.120140
17	[Type, TS, Union]	119	0.107494
18	[Type, TS, ObjectMember]	102	0.092138
19	[Git, Change, Copy]	102	0.092138
20	[NPM, Script]	91	0.082201
21	[TS, Property]	65	0.058715
22	[TS, Function]	47	0.042456
23	[Type, TS, FunctionParameter]	40	0.036132
24	[Git, Branch]	40	0.036132
25	[Type, Object, TS]	39	0.035229
26	[File, Directory]	34	0.030713
27	[Type, TS, Function]	34	0.030713
28	[TS, Parameter]	33	0.029809
29	[Package, File, Json, NPM]	29	0.026196

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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Nodes per label combination (more than 0.5% overall)

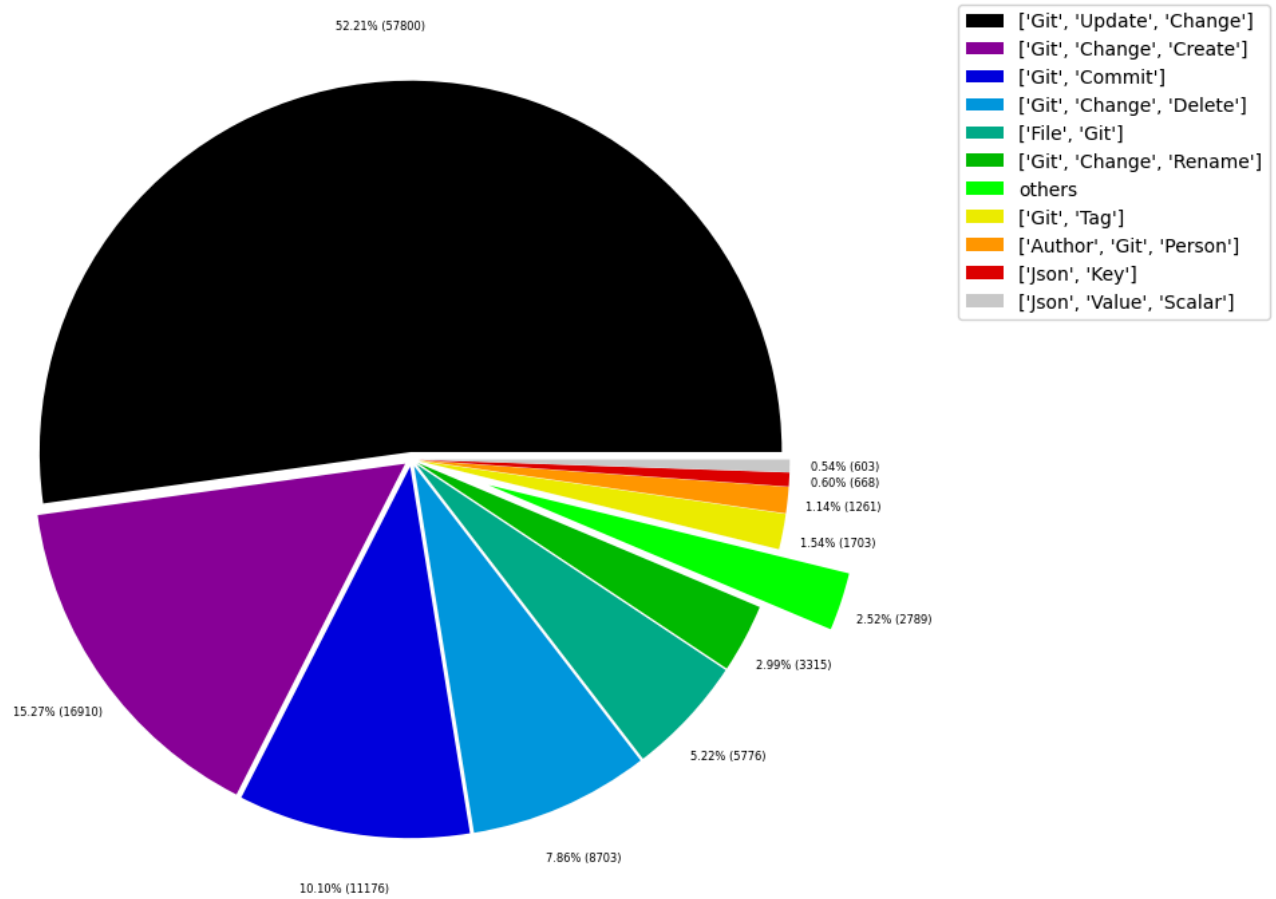


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000903
1	[File, TS, Scan]	1	0.000903
2	[TS, Method]	1	0.000903
3	[Repository, File, Git]	1	0.000903
4	[TS, Constructor]	1	0.000903
5	[Value, TS, ObjectMember]	1	0.000903
6	[TS, Class]	1	0.000903
7	[TS, Enum]	2	0.001807
8	[Value, Object, TS]	3	0.002710
9	[Type, TS, Tuple]	3	0.002710
10	[Value, TS, Function]	4	0.003613
11	[TS, TypeParameter]	4	0.003613
12	[Value, TS, Complex]	5	0.004517
13	[NPM, Engine]	6	0.005420
14	[File, Local]	6	0.005420
15	[Project, TS]	6	0.005420
16	[Type, TS, TypeParameterReference]	6	0.005420
17	[Value, TS, Call]	6	0.005420
18	[Value, TS, Member]	6	0.005420
19	[File, TS, Local, Module]	6	0.005420
20	[TS, EnumMember]	8	0.007226
21	[Type, TS, NotIdentified]	11	0.009936
22	[TS, ExternalModule]	11	0.009936
23	[Json, Value, Array]	12	0.010840
24	[Value, TS, Declared]	13	0.011743
25	[TS, TypeAlias]	16	0.014453
26	[File, Directory, Local]	16	0.014453
27	[TS, Interface]	17	0.015356
28	[Type, TS, Intersection]	17	0.015356
29	[JQAssistant, Rule, Concept]	19	0.017163

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

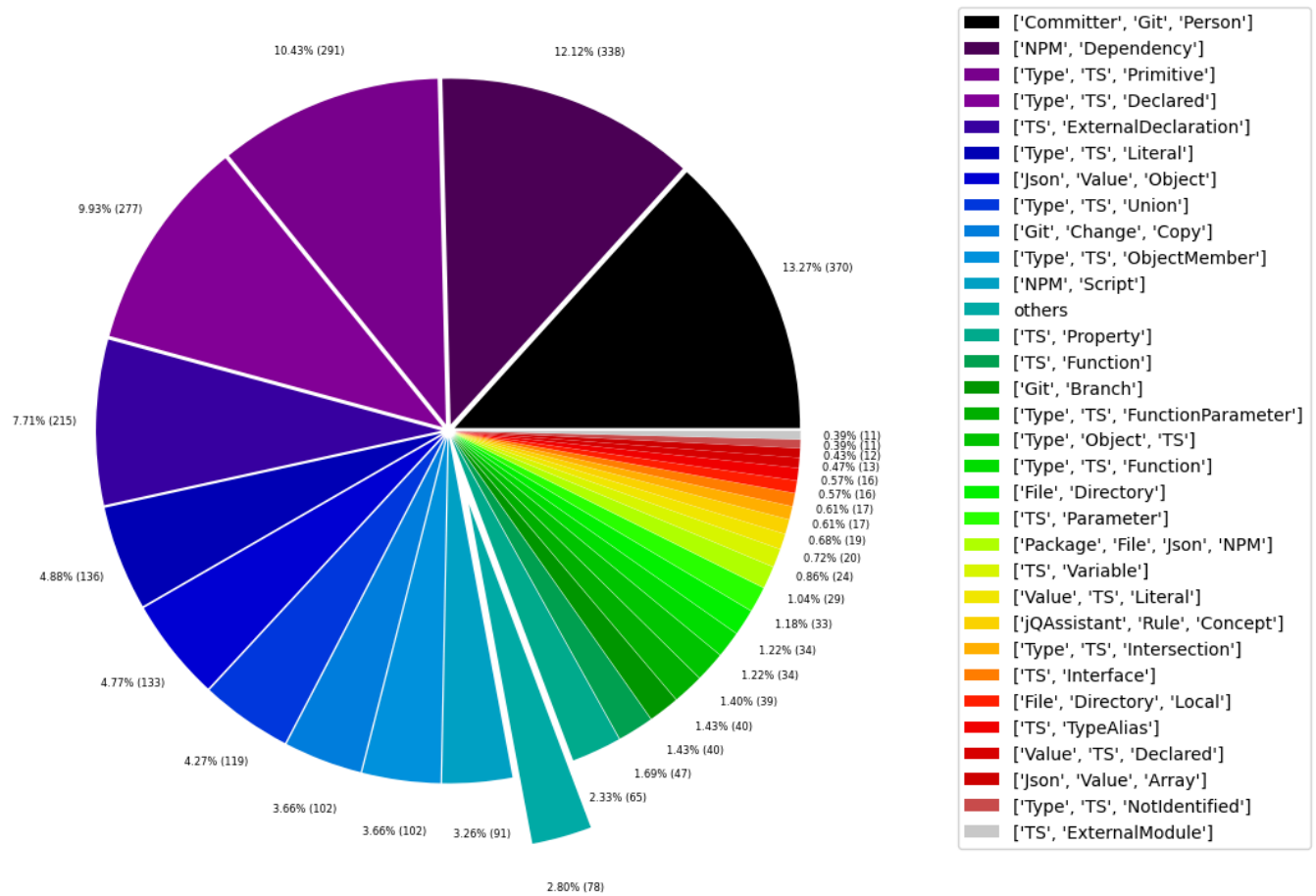


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

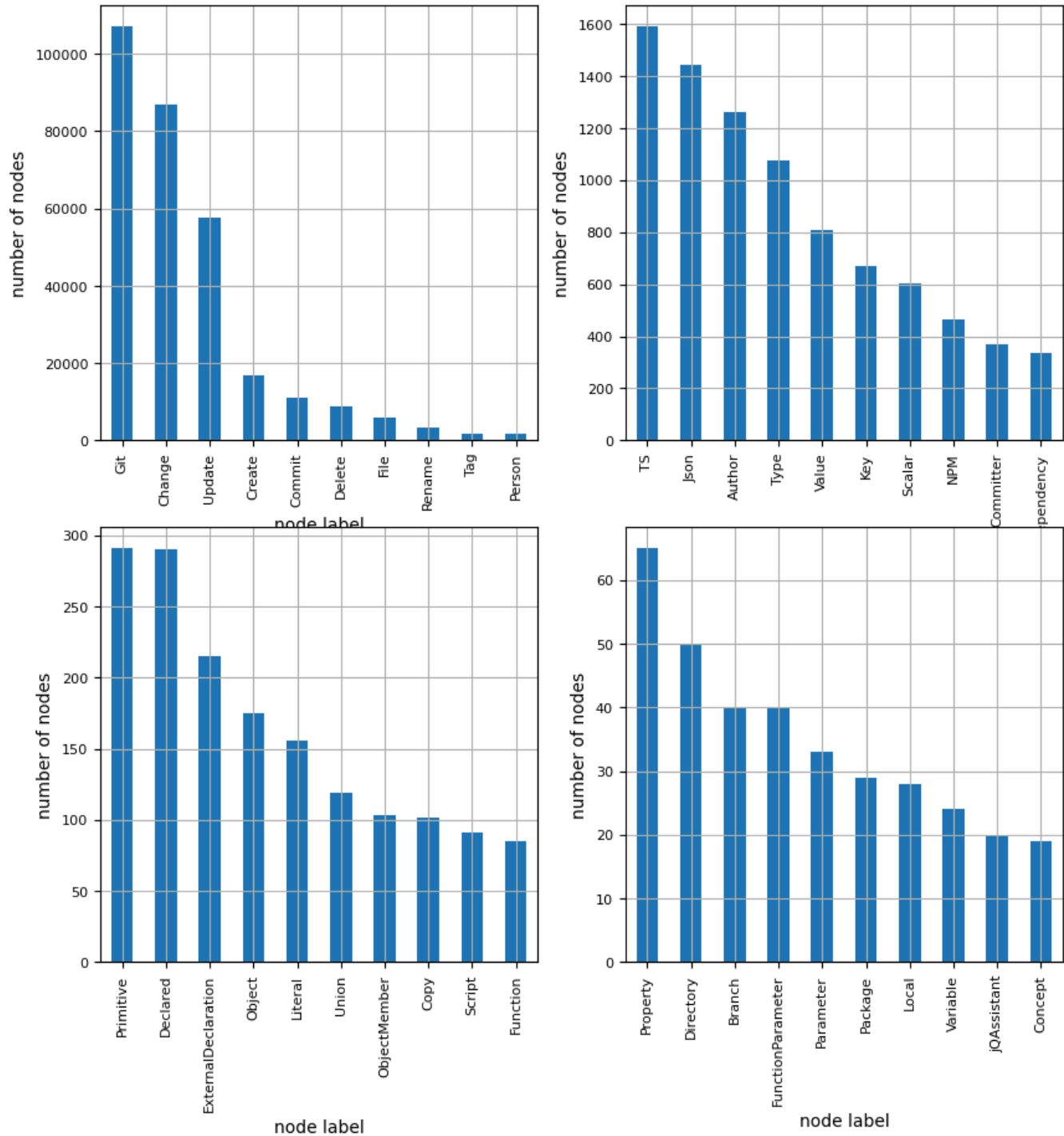
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	107157	96.795960
1	Change	86830	78.434384
2	Update	57800	52.211302
3	Create	16910	15.274967
4	Commit	11176	10.095390
5	Delete	8703	7.861505
6	File	5869	5.301525
7	Rename	3315	2.994472
8	Tag	1703	1.538336
9	Person	1631	1.473298
10	TS	1591	1.437166
11	Json	1445	1.305283
12	Author	1261	1.139074
13	Type	1075	0.971058
14	Value	806	0.728068
15	Key	668	0.603411
16	Scalar	603	0.544696
17	NPM	464	0.419136
18	Committer	370	0.334225
19	Dependency	338	0.305319
20	Primitive	291	0.262863
21	Declared	290	0.261960
22	ExternalDeclaration	215	0.194212
23	Object	175	0.158079
24	Literal	156	0.140916
25	Union	119	0.107494
26	ObjectMember	103	0.093041
27	Copy	102	0.092138
28	Script	91	0.082201
29	Function	85	0.076781
30	Property	65	0.058715
31	Directory	50	0.045165
32	Branch	40	0.036132
33	FunctionParameter	40	0.036132
34	Parameter	33	0.029809
35	Package	29	0.026196
36	Local	28	0.025293
37	Variable	24	0.021679
38	jqAssistant	20	0.018066
39	Concept	19	0.017163

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count` .

Total number of relationships: 330622

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	86830	26.262620
1	MODIFIES	86830	26.262620
2	UPDATES	57800	17.482200
3	COMMITTED	22352	6.760591
4	CREATES	20327	6.148109
5	HAS_PARENT	12251	3.705440
6	DELETES	12018	3.634967
7	HAS_COMMIT	11176	3.380295
8	HAS_FILE	5776	1.747010
9	RENAMES	3315	1.002656
10	HAS_NEW_NAME	1763	0.533237
11	HAS_TAG	1703	0.515090
12	ON_COMMIT	1703	0.515090
13	HAS_AUTHOR	1261	0.381402
14	DEPENDS_ON	887	0.268282
15	HAS_KEY	668	0.202043
16	HAS_VALUE	668	0.202043
17	CONTAINS	594	0.179661
18	HAS_COMMITTER	370	0.111910
19	OF_TYPE	338	0.102232
20	EXPORTS	309	0.093460
21	REFERENCES	197	0.059585
22	DECLARES	186	0.056258
23	DECLARES_DEV_DEPENDENCY	169	0.051116
24	DECLARES_DEPENDENCY	161	0.048696
25	HAS_MEMBER	103	0.031153
26	COPIES	102	0.030851
27	HAS_TYPE_ARGUMENT	94	0.028431
28	DECLARES_SCRIPT	91	0.027524
29	RETURNS	82	0.024802

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

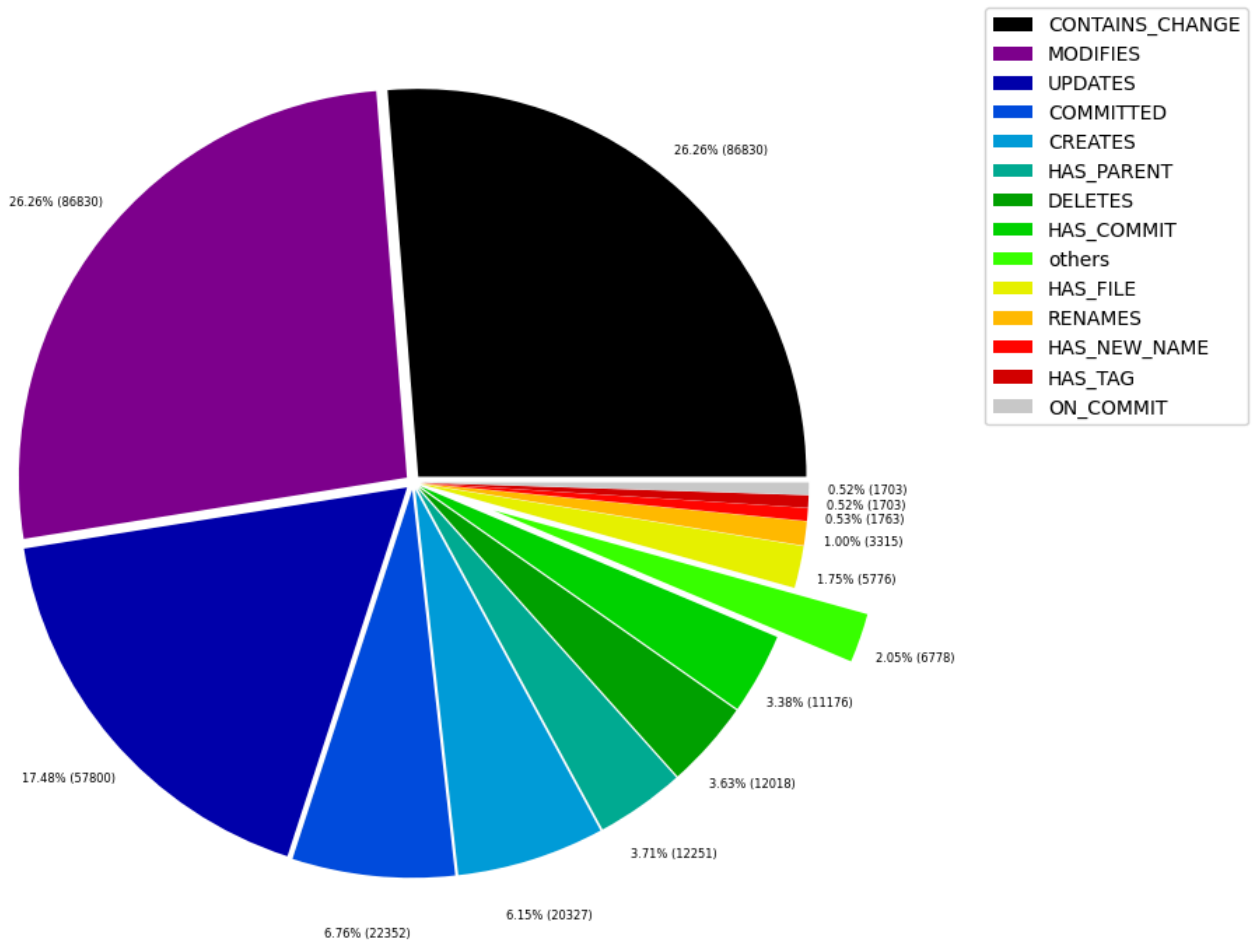


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

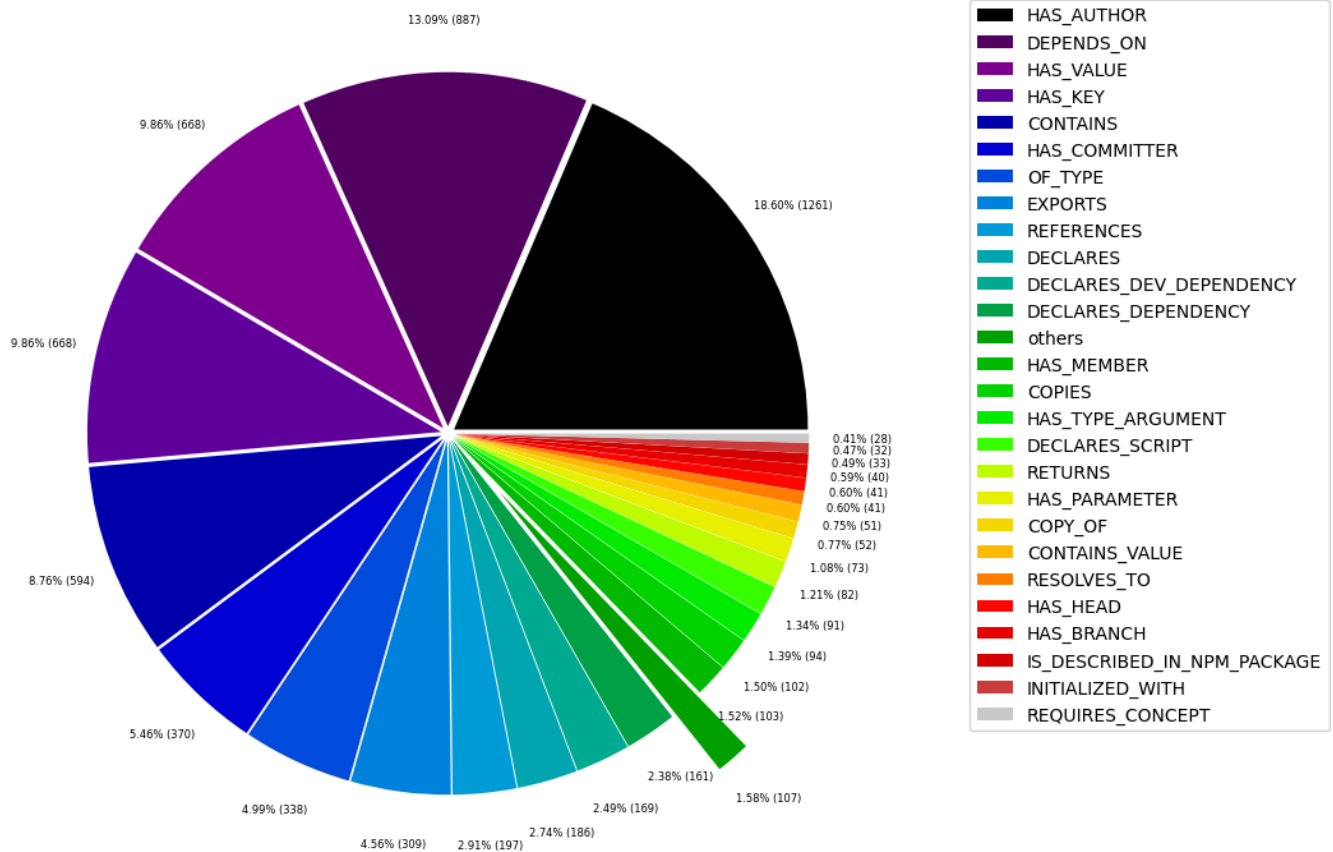
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONSTRAINED_BY	4	0.001210
1	REFERENCED_PROJECTS	5	0.001512
2	MEMBER	6	0.001815
3	HAS_ROOT	6	0.001815
4	HAS_NPM_PACKAGE	6	0.001815
5	HAS_CONFIG	6	0.001815
6	HAS_ARGUMENT	6	0.001815
7	EXTENDS	6	0.001815
8	DECLARES_ENGINE	6	0.001815
9	CONTAINS_PROJECT	6	0.001815
10	CALLS	6	0.001815
11	PARENT	6	0.001815
12	DECLARES_PEER_DEPENDENCY	8	0.002420
13	USES	11	0.003327
14	INCLUDES_CONCEPT	19	0.005747
15	REQUIRES_CONCEPT	28	0.008469
16	INITIALIZED_WITH	32	0.009679
17	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009981
18	HAS_BRANCH	40	0.012098
19	RESOLVES_TO	41	0.012401
20	HAS_HEAD	41	0.012401
21	CONTAINS_VALUE	51	0.015425
22	COPY_OF	52	0.015728
23	HAS_PARAMETER	73	0.022080
24	RETURNS	82	0.024802
25	DECLARES_SCRIPT	91	0.027524
26	HAS_TYPE_ARGUMENT	94	0.028431
27	COPIES	102	0.030851
28	HAS_MEMBER	103	0.031153
29	DECLARES_DEPENDENCY	161	0.048696

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	57800	57800	
1	[Git, Update, Change]	UPDATES	[File, Git]	57800	57800	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	57800	11176	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	16910	11176	
4	[Git, Change, Create]	MODIFIES	[File, Git]	16910	16910	
5	[Git, Change, Create]	CREATES	[File, Git]	16910	16910	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12251	11176	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11176	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11176	1261	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11176	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8703	11176	
11	[Git, Change, Delete]	DELETES	[File, Git]	8703	8703	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	8703	8703	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5776	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3315	11176	
15	[Git, Change, Rename]	DELETES	[File, Git]	3315	3315	
16	[Git, Change, Rename]	RENAMES	[File, Git]	3315	3315	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3315	3315	
18	[Git, Change, Rename]	CREATES	[File, Git]	3315	3315	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1763	5776	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1703	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1703	1703	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1261	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

total_number_of_nodes (vertices): 110704

total_number_of_relationships (edges): 330622

-> total directed graph density: 2.6977956198748876e-05

-> total directed graph density in percent: 0.0026977956198748878